Lab 39: Switch off the one noisy category, keep the accurate ones

Topic 4: Prompt Engineering & Structured Output · Task 4.1: Design prompts with explicit criteria to improve precision and reduce false positives

Runs on Simulator 3: Multi-Agent Research System · Self-practice

Source: SkillCertPro practice exam Part 1, Q20 (unofficial third-party question)

Commit to an answer first

This team added a prompt-based quality gate that classifies draft report findings before publication. One category, “unsupported market claim”, now produces many false positives on well-cited synthesis paragraphs, while other categories, such as “missing citation” and “source conflict”, are performing well. Engineers have started ignoring the entire quality report. What should the team do next?

A. Temporarily omit the noisy category from the active prompt, keep accurate categories enabled, and rebuild it with examples, definitions, and precision/recall evals.

B. Force the quality gate to return strict JSON, reject malformed outputs, and use schema compliance as the main trust metric.

C. Disable every quality category for a release cycle, collect no automated findings, and re-enable all categories after prompt edits.

D. Keep the noisy category active, add a warning that false positives are expected, and ask engineers to manually disregard those findings.

Simulate the solution

```
# In the browser: open Simulator 3, then Lab 39
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 39

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py --case UC-20
```

What you should see

You write the revised category list and an eval plan with a measurable threshold.

A revised category list plus an eval plan with a measurable threshold.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

The answer

A. Omit the noisy category, then rebuild it with examples and evals.

Separate the noisy signal from the useful one. Categories are application-defined lists in the prompt, so suppressing one is an application-level control, not a built-in toggle. Don't let it disappear for good, though: treat it as a classification problem. Collect examples that should and should not be flagged, including edge cases, define the category precisely, and evaluate with precision, recall and F1. Strict JSON helps parsing but cannot make the judgement correct.

Why the others fail

B. Schema compliance measures format, not whether a flag is right.

C. Throws away the categories that are working and the trust they earn.

D. Leaves the noise in place, and with it the erosion of trust in the whole report.

Check yourself

- noisy category removed from the active prompt only
- accurate categories remain enabled
- re-enable gate is precision/recall on labelled examples, not schema compliance

Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.